

CHAPTER 14

Calendar Spread Trading

Time-based spreads and the term structure — trading the shape of the forward curve, from contango and backwardation to the Brent complex.



SAME BARREL, DIFFERENT MONTH

Time itself has a price

The same product for delivery next month rarely costs the same as for delivery in six. That difference — the calendar spread — is the market's price on holding a barrel through time. Its pattern across all months is the **term structure**, and it can be traded entirely on its own.

A spread between months

The price for one delivery month minus another — December crude minus June crude — of the very same grade.

It prices storage & scarcity

The shape of the forward curve tells you whether the market wants barrels stored now or delivered now.

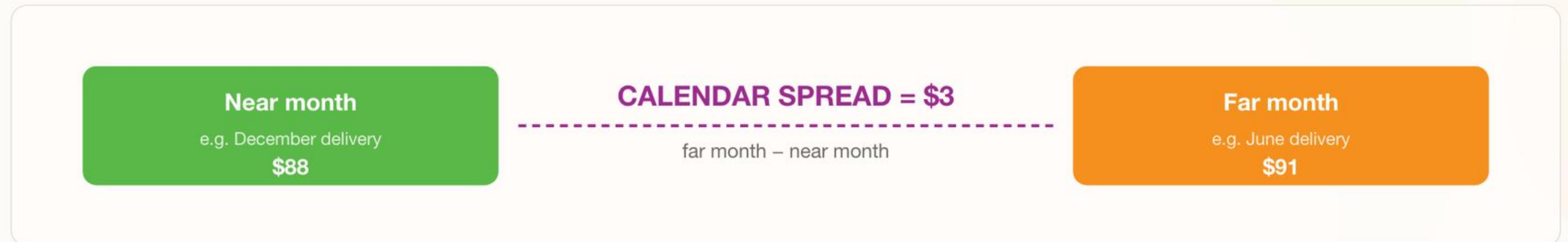
A bet on the slope

You don't trade the level of oil — you trade whether the curve will steepen or flatten between two dates.

THE DEFINITION

What a calendar spread is

The price of one delivery month minus another for the same product — the value of time, with grade and location held constant. Buy "X/Y" means buy X and sell Y.



Quoted as one number. A "Dec–Jun" of +\$3 means June trades \$3 over December — and you trade that figure, not either outright price. (A "Jan–Feb" with a dash is instead a strip — the average of both months.)

THE SHAPE OF THE CURVE

Contango vs backwardation

Oil curves cannot be explained by the time value of money alone — their shape is driven by supply-and-demand expectations. They take one of two forms.

Contango

far above near — well supplied



Future months cost more. The market pays you to wait — and rewards anyone who can store.

Backwardation

near above far — tight market



Near months cost most. Barrels are wanted now; holding for later loses money — the signature of scarcity.

PROMPT VS SUSTAINABLE PRICE

Why the curve takes its shape

The front of the curve — the "prompt" price — moves around a slower-changing view of oil's long-term sustainable value. Which side it sits on dictates how buyers and sellers behave.

Contango — prompt is cheap

Why: Prompt over-supply, high stocks, or expected future tightness (outages, declining production).

The front sits below the long-term view, so BUYERS grab all they can at today's low price; SELLERS hold back and sell as little as possible.

Backwardation — prompt is dear

Why: Prompt under-supply — an unscheduled refinery outage, a cold snap, low stocks, or expected future supply growth.

The front sits above the long-term view, so BUYERS take only the minimum they need; SELLERS push out as much as they can while prices are high.

WHY THE SHAPE IS BOUNDED — ON ONE SIDE

Storage economics & the asymmetry

The curve is the market pricing the cost and value of holding a barrel over time — and the two directions are not symmetric.

Contango rewards storing

Buy cheap today, store, sell forward dear — so the curve bids up storage. Inventories build and the contango deepens.

Backwardation penalizes storing

When near months pay most, holding means selling later for less. The market shouts "deliver now, don't store".

Contango is capped

It can only widen until cash-and-carry storage arbitrage kicks in. Once that's exhausted — tanks full — even contango has a ceiling.

Backwardation is (almost) unlimited

There is no such ceiling on the upside: a consumer facing a supply cut-off will pay almost any prompt price, so backwardation can run wild.

THE TERM-STRUCTURE FRAMEWORK

Cost of carry vs convenience yield

Beneath every curve sits a tug-of-war between the cost of holding a barrel and the benefit of having it on hand. Whichever wins sets the shape.

Cost of carry

storage + insurance + financing (interest)
higher carry → pushes toward CONTANGO

VS

Convenience yield

value of having product on hand vs force majeure
higher yield → pushes toward BACKWARDATION

Other drivers fine-tune it: interest and inflation rates, liquidity and volatility that vary along the curve, and the fact that **long-dated contracts move less than the front** — so heavy hedging on the back end can reshape the whole structure.

THE STORAGE TRADE

Cash-and-carry — trading contango

When the curve is in contango, the spread can be captured physically: buy a barrel now, store it, and deliver it against a higher forward price.



This is exactly why contango is capped: profit exists only while **spread > storage cost**, and the very buying-and-storing that captures it flattens the curve back into line.

LIVING ON THE CURVE

The roll and roll yield

A trader who holds a position past expiry must "roll" it to the next month — and the curve's shape decides whether that roll quietly earns or bleeds money.

Rolling in backwardation earns

Sell the expiring near month high, buy the next month cheaper — a positive roll yield that compounds over time.

Rolling in contango costs

Sell low, buy the next month dearer — a negative roll yield that quietly erodes a long position month after month.

It can dwarf the flat move

For long-term holders, roll yield often matters more than spot — a flat market can still lose money in deep contango.

The same trade, on purpose

A calendar-spread trader takes the roll deliberately — long the month they expect to gain, short the one they expect to lag.

PUTTING ON THE TRADE

Trading the slope of the curve

You buy or sell a calendar spread because you have a view on the **slope** of the forward curve — not the flat price.

Buy the spread → toward backwardation

Buy near, sell far. You profit if the front strengthens relative to the back — the curve tilts into (or deeper) backwardation.

Sell the spread → toward contango

Sell near, buy far. You profit if the front weakens relative to the back — the curve tilts into (or deeper) contango.

What it's used for

Hedging buy/sell pricing-window mismatches, speculation on the curve, and pricing the building blocks of long-dated swaps.

How it's booked

Quoted as a single differential, but often entered as two separate monthly deals once executed — one leg per month.

THE CALENDAR WITHIN THE CALENDAR

Seasonal calendar spreads

Some spreads are driven by the seasons — winter heating and summer driving leave a predictable footprint on the curve.

Winter distillate, summer gasoline

Heating-oil and diesel months over winter, and gasoline months over summer, tend to trade at a premium to the shoulders.

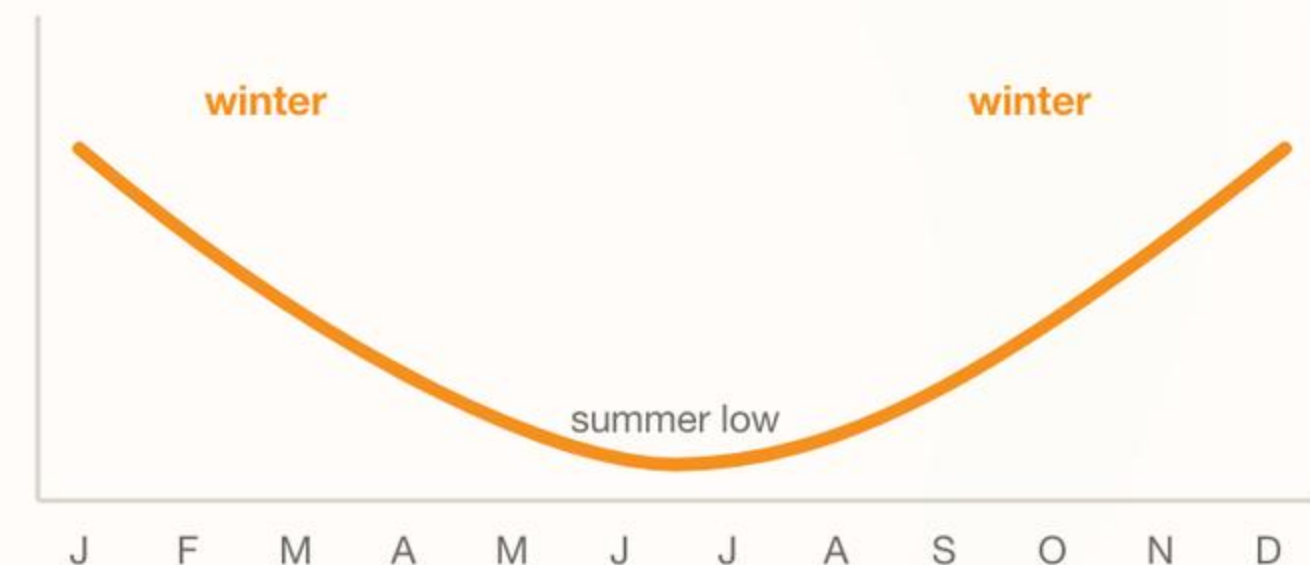
A repeatable footprint

Because the pattern recurs, traders position calendar spreads ahead of the season — buying the strong month, selling the weak.

Not a sure thing

A mild winter or a demand shock can flatten the seasonal premium — the pattern shapes odds, it doesn't guarantee them.

HEATING OIL — SEASONAL PRICE SHAPE



HOW IT'S TRADED IN PRACTICE

The Brent term-structure complex

The world's main crude benchmark is a ladder of linked instruments — physical at one end, paper at the other, connected so the whole structure can be priced and hedged.

1 Dated Brent (DTD)

PHYSICAL · SPOT

Platts' daily 16:30 assessment of wet BFOE cargoes loading ~10–21 days forward. The physical spot benchmark.



2 21-Day BFOE (cash)

FORWARD CASH · MONTHS

Forward cash contract (21 days–3 months) quoted by month — effectively the forward price of Dated Brent, with physical delivery.



3 Brent Futures (ICE)

PAPER · UP TO YEARS

Exchange-traded paper futures on ICE, linked back to the physical via EFP (exchange for physical) and 21-Day BFOE.

The gaps *between* these rungs are themselves traded as swaps — that's where CFDs and DFLs come in.

TRADING THE GAPS IN THE STRUCTURE

Brent CFDs and DFLs

Two short-term swap markets let traders take a view on the very front of the Brent term structure without touching a cargo.

CFD

Contract for Difference

Dated Brent vs forward cash BFOE

Short-term swaps quoted for each of the eight weeks ahead, against the second forward BFOE cash month. They price the gap between physical Dated Brent and near-forward cash — the steepness of the front of the curve.

DFL

Dated-to-Frontline

Dated Brent vs ICE front-month future

Swaps on the difference between the daily Platts Dated Brent assessment and the ICE front-month futures settlement. Quoted six months, six quarters and three years out — the physical-to-paper basis.

Together they let a trader hedge or speculate on **exactly where physical Brent sits relative to the forward and futures curve** — the practical core of term-structure trading.

KNOW THE DOWNSIDE

The risks of calendar trading

A calendar spread is "lower risk" than a flat-price bet — but lower is not zero, and the curve can turn fast.

The curve can flip

Contango can swing to backwardation (or back) on a shock, moving the spread sharply against an unprepared position.

Storage & financing costs

A physical cash-and-carry only profits if real storage, insurance and financing stay below the spread you locked in.

Front-month volatility

The nearest contract can whip around at expiry, so the front leg of a spread is often the most dangerous.

Still leveraged

Each leg is a margined position — a "small" spread move is amplified, so risk discipline still applies.

CONCLUSION — TRADING THE TERM STRUCTURE

The shape of the curve is a tradable view.

Shape over level

Contango rewards storage; backwardation signals scarcity — and you trade the slope, not the flat price.

Carry vs convenience

The curve balances the cost of holding a barrel against the value of having it on hand.

A real toolkit

From cash-and-carry to Brent CFDs and DFLs, the term structure is traded with a precise set of instruments.